# Machine-Checked Foundations for the Exact-Rational Riccati Certificate: A Lean Companion Note

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Version 1 — August 2026

## 1 What this note is

This note is the verification companion to the paper on exact-rational certification of block-banded trajectory QCQPs [2], and it does three jobs. It documents the Lean 4 formalization behind the machine-checked results cited there: each such citation resolves to the numbered section here that states the corresponding declaration and its hypotheses. It catalogues the executable receipts behind the paper's measured and instance-level results (§6). It records the mechanization of the published Podium technical note's theorems (§7).

The corpus machine-checks classical facts along the paper's trusted path rather than new mathematics: what it supplies is a tolerance-free, mechanically verified chain, not novelty in the individual lemmas.

## 2 The corpus and its pin

The trusted path is mechanized in eighteen top-level Lean 4 files, all `sorry`-free, in the Lean corpus of the Podium project (`github.com/adi-oltean/podium`). Every declaration this note cites is named explicitly, so a reader can locate it by name in any copy of the corpus. The formalization is also available from the author at `adi@starcloud.com`.

Every file type-checks against a pinned project built from `leanprover/lean4:v4.33.0` and Mathlib at revision `v4.33.0`. A per-file regression script drives the build: it reports pass or fail for each file and surfaces `#print axioms` for the headline declarations. The axiom audit admits only `propext`, `Classical.choice` and `Quot.sound`, and never `sorryAx`.

## 3 The acceptance predicate, both directions

`ldlt_isPSD` (`LDLTSound.lean`): if $M = L^\top \operatorname{diagonal}(d)\, L$ and every pivot $d_i \geq 0$, then $M$ is PSD in the quantifier sense $\forall v,\ 0 \leq v^\top M v$. This is the acceptance direction; it holds at any order and over any linear ordered field, by the expansion $v^\top M v = \sum_i d_i (Lv)_i^2$.

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`ldlt_complete` (`SchurStep.lean`): a symmetric PSD matrix of any order $m$ admits weights $d_j \geq 0$ and vectors $v_j$ with $M = \sum_j d_j v_j v_j^\top$. The induction peels one nonnegative-weighted rank-one term per pivot, and it passes through singular pivots because a zero pivot forces a zero border (`headBordered_border_zero`). This is the converse of `ldlt_isPSD` up to padding the factor to square; the residual difference is purely dimensional ($k \leq m$).

`isPSD_iff_ldlt` (same file) packages the two directions as one equivalence: for symmetric $M$, $M$ is PSD if and only if such a nonnegative-weight decomposition exists. This is the machine-checked form of accept $\Leftrightarrow$ PSD, with *accept* being exactly that existential.

### 3.1 The bridge to the library-standard notion

`posSemidef_iff_isPSD` (`PosSemidefBridge.lean`): for a rational $n \times n$ matrix $M$, Mathlib's `Matrix.PosSemidef` $M$ holds exactly when $M^\top = M$ and `IsPSD` $M$, the quantifier form $\forall v,\ 0 \leq v^\top M v$. Symmetry is derived by the equivalence, not assumed of $M$. So the quantifier form is the library-standard notion rather than a bespoke one. That bridge is compiled on its own and is not linked inside Lean to the predicate the acceptance lemmas above are stated against; the condition is stated in §8.

## 4 Certificate soundness and closure

`general_sprocedure_sound` (`GeneralLifting.lean`) takes three hypotheses:

1. the bordered matrix built from Hessian $P$, half-linear part $b$ and corner $c$ is PSD;
2. every multiplier $\lambda_i \geq 0$;
3. the form identity: for every point, the bordered quadratic form at the lift equals $f_0 - \sum_i \lambda_i f_i - t$.

From these it concludes $t \leq f_0(y)$ at every $y$ with all $f_i(y) \geq 0$. The theorem holds at any order $n$, at any number $m$ of constraints, and over any linear ordered field.

Of those three, the form identity is the one not mechanized in general. The abstract lifting identity `bordered_lifting` (same file) is proved at general $n$: the bordered matrix's quadratic form at $(x, 1)$ equals $x^\top P x + 2b^\top x + c$. Its instantiation to a specific S-procedure assembly is discharged per instance by `ring`, and the capstone theorems carry that instantiation as a hypothesis too.

`certified_global_optimum` (`ClosureOptimum.lean`): given multipliers $\lambda \geq 0$, a witness $\bar{x}$ that minimizes the Lagrangian $f_0 - \sum_i \lambda_i f_i$ over all points, and complementary slackness $\sum_i \lambda_i f_i(\bar{x}) = 0$, the witness satisfies $f_0(\bar{x}) \leq f_0(y)$ at every feasible $y$ — the attainment leg, over any linear ordered field. Its companion `bracket_exact` (same file) adds feasibility of $\bar{x}$ and a supplied lower bound — that $t \leq f_0$ on the feasible set — and concludes both $t \leq f_0(\bar{x})$ and global optimality of $\bar{x}$. The equality $t^\star = f_0(x^\star)$ that closes the bracket is by construction of $t^\star$ and is not separately mechanized.

`certified_optimum_pipeline` (`Certificate.lean`) assembles the chain as one theorem. Its hypotheses are an accepted exact factorization $M = L^\top D L$ with pivots $\geq 0$, multipliers $\lambda \geq 0$, the form identity, and a feasible Lagrangian-minimizing witness $\bar{x}$ with complementary slackness. It returns three conclusions: $t$ lower-bounds $f_0$ on the feasible set; $\bar{x}$ is a global optimum; and $t \leq f_0(\bar{x})$. From these, $f_0(\bar{x}) = J^\star$ follows because $\bar{x}$ is feasible and minimal. Two fully worked exact-$\mathbb{Q}$ instances, a one-dimensional keep-out and a two-dimensional disk, run through the verdict-level twin `certified_optimum_of_verdict`, which starts from the PSD verdict rather than from a factorization.

## 5 The diagonal-dominance leg

`isPSD_of_wdd` (`GershgorinPSD.lean`): symmetry together with row dominance $\sum_{j \neq i} |M_{ij}| \leq M_{ii}$ implies PSD at general $n$ over any linear ordered field, and the declaration takes exactly those two hypotheses. This is the Gershgorin leg of the paper's diagonal-dominance lemma.

## 6 The executable receipts

The paper's measured and instance-level results rest on a set of executable receipts: exact-arithmetic Python scripts that construct the instances, run them through the trusted checkers, and assert what the paper reports. This section catalogues those receipts by role. Every receipt below ships in the paper's supplementary bundle, together with the supporting modules the drivers import, with one exception: `test_riccati` is part of the Podium library's own test suite. The bundle's own `README.md` gives the exact setup and invocation for every receipt below: the library commit to clone, the dependencies, and the command line for each script and for the pytest suite. Throughout the receipts, the bit-size of a rational is `numerator.bit_length()` + `denominator.bit_length()`.

The trusted checkers are not bundle contents: they live in the Podium repository at `github.com/adi-oltean/podium`, and the receipts import them from there (the library import requires NumPy). The band checkers are `border_band_psd` and `block_tridiag_psd` in `podium.verify.riccati`; retrieving them together with `test_riccati` requires commit `c96fcb5` of that repository. The archived Zenodo release of the library, v0.8.1 (DOI 10.5281/zenodo.21302775), predates `podium.verify.riccati`: it carries `podium.verify.bracket` (the dense $M$-test) and `podium.verify.barrier`, but not the band sweeps.

**The family drivers.** Three drivers construct the families of the paper's results table in exact `Fraction` arithmetic, and each verifies its emitted certificate through the exact dense $M$-test. The trusted checkers refuse floats; the CW state-transition matrix alone is synthesized in floating point and rationalized before any certification.

`shared_state_exact.py` constructs the scalar shared-state family; it pins the results table's scalar row.

`vector_state.py` constructs the vector shared-state family ($d = 1..4$) and pins the table's vector row. Its probe `gap_probe` ($\lambda_k > w_k$ at $N$=4, $d$=2) is the aggregate-PSD-violated instance of the table's refusal row, on which the checker returns no certificate.

`rpod_cw.py` constructs the CW-coupled keep-out family of the paper's demonstration section and pins the table's CW row. The family's data are mean motion $n_0 = 10^{-3}$ rad/s, sampling phase $\tau = n_0 \Delta t = 1/2$, weights $R = I_4$ and $W_k = w_k I_4$, per-stage radii cycling $\{1/2, 1, 3/2\}$, and STM denominators capped at $10^5$. The driver also prints the block pivots $S_k$, which grow measurably linearly in the horizon: 780 bits at $N$=2 to 17889 at $N$=24.

**The measurement engines.**

`cert_metrics.py` recomputes both bit columns of the results table for the three closing families at the headline sizes, and re-verifies each of those rows through all three trusted routes: the dense $M$-test, the band-$H$ sweep, and the bordered-$M$ sweep.

`cw_anatomy.py` recomputes, at all 23 horizons $N = 2, \dots, 24$, every denominator statistic the paper quotes for the CW-coupled family, together with the closure verdicts and both bit columns. It asserts the divisibility law behind the certified value's denominator, and the value's size inequalities, at every horizon. It measures the entrywise rationalization error of the CW state-transition matrix. Its bordered-sweep section pins the corner-carrying sweep pivot by pivot against the dense $LDL^\top$ of $M$ at the shortest horizons, which is the test-pinning leg of the paper's bordered-extension remark.

**The exact instance receipts.** Four receipts certify the paper's named instances entirely over $\mathbb{Q}$; the fourth certifies two sibling instances in one run.

`single_keepout_coupling_gap.py` is the paper's certified duality-gap example. It asserts $J^\star = 212/13$ by exact enumeration of the branch-KKT candidates, certifies through a rational moment point that every level-one certificate is capped at $815/64$, and certifies the level-one gap $\geq 2973/832$. The same receipt discharges the diagnosis: stationarity recovers $\lambda^\star = 24/13$, and $H(\lambda^\star)$ has determinant $-7/4$, so aggregate-PSD fails there.

`two_keepout_stage.py` closes the bracket at $t^\star = 169/320$ on a stage with two active keep-outs whose contacts touch disjoint coordinates; this is one half of the main theorem's multi-keep-out generality. Its published certificate is also the paper's floating-point counterexample object: the same sweep transcribed into IEEE-754 doubles refuses it, the terminal corner coming out $-4.44 \times 10^{-16}$ rather than exactly 0.

`overlap_keepout_stage.py` closes the bracket at $t^\star = 5773/3600$ on a stage whose two active contact gradients share their whole support; this is the other half of the multi-keep-out generality, and the receipt checks that the verifier's normal-equation solve recovers $\lambda^\star$ on exactly that stage.

`hard_dynamics_gap.py` certifies both halves of the paper's hard-dynamics discussion over $\mathbb{Q}$, on $d{=}1$ instances with one keep-out and one equality. On the first instance the best constant-multiplier bound is exactly 1 against $J^\star = 3/2$, a certified gap of $1/2$ whose diagnosis is an aggregate-PSD failure. On its sibling the aggregate-PSD analogue holds and the constant-multiplier bracket closes exactly at $t^\star = J^\star = 9/4$.

**The pytest pins.**

`test_bracket_scaling.py` ships in the bundle and pins the receipts above: it executes each of the four exact instance receipts and asserts its published constants, runs `cw_anatomy.py`, and checks `cert_metrics.py`'s results-table rows at the headline sizes.

`test_riccati` ships in the Podium library and pins the band checkers themselves: the band-$H$ sweep on the assembled $H$ and the bordered sweep on the assembled $M$ are checked against the dense reference predicate `is_psd` of `podium.verify.barrier`, on randomized fixed-seed instances plus dedicated singular-$H$ and zero-pivot cases.

**The height receipts.** These receipts pin the paper's size measures and the separations among its named heights.

`height_measure_check.py` pins the paper's height and bit-size definition: the enumerated least height is attained at the common-denominator form over the lcm of the reduced denominators, per-entry bit-size obeys bit $\le 2\,\mathrm{ht}$, and a positive integer $D$ read as a rational obeys $\mathrm{bit}(D) = \mathrm{bitlen}(D) + 1$.

`heights_enumerated.py` asserts that on the CW-coupled family the four height-bearing tuples carry four pairwise distinct heights at every one of the 23 horizons, in the strict order $\mathrm{ht}(x^\star) < L < L_M < L_{\mathrm{asm}}$.

`height_inversion.py` exhibits a family admissible under the paper's constructed-family definition on which the assembled height $L_{\mathrm{asm}}$ is the *smaller* of the two design-side heights — the receipt behind the statement that the definition forces no ordering between $L$ and $L_{\mathrm{asm}}$.

`height_governance.py` exhibits an admissible family, closing on all three trusted checkers, with $L_M \le 8$ bits against a design height $L = 257$–$258$, so that there the per-stage design height bounds every entry of the bordered matrix — the ordering opposite to the CW family's.

`claim311_recheck.py` extends the certified-value measurements past the demonstration window by an exhaustive exact sweep of every horizon $N = 2, \ldots, 6143$: $\mathrm{den}(t^\star)$ stays inside 450–526 bits at every horizon, while $\mathrm{bit}(t^\star)$ keeps the residual $\log N$ drift.

`claim258_const_lm_witness.py` constructs the constant-$L_M$ witness: an admissible closing family with $L_M = 2$ at every horizon whose largest interior pivot grows without bound, so no function of $L_M$ alone bounds the sweep's cost.

`attackb_scope.py` constructs the fixed-order witness — at order $n+1 = 9$, multipliers of growing denominator raise $L_M$ and the largest interior pivot together, so no function of the order alone bounds the cost either. It also separates the multiplier-recovery solve's operand heights from its formation heights on the demonstration and inverted families.

**`attack2_scope.py` and `attackb_scope.py`** measure the recovery solve's operands at the design height: 5–6 bits against $L = 6$ on the CW family (`attackb_scope.py`), and 259 bits against $L = 258$ on `height_inversion.py`'s inverted family (`attack2_scope.py`).

**`claim235_thirdcheck.py` and `claim306_attack.py`** jointly pin the two witnesses for why the certified value's bit bound carries a horizon term and is not stated against $L$ alone: an admissible closing family with $L_{\mathrm{asm}} = 2$, $\mathrm{ht}(x^\star) = 1$ and $t^\star = N$ exactly, and integer goal weights of growing magnitude taking $\mathrm{den}(t^\star)$ unbounded at fixed $L = 1$.

**The class-structure and sweep-scope receipts.** These receipts pin structural facts about the class and the scope of the band sweep.

`bandwidth_scope.py` pins the scalar bandwidth: the neighbour-only coupling confines every nonzero entry to half-width $b = 2d-1$, attained whenever some coupling block has a nonzero $(1, d)$ entry, while structured couplings such as $A_k = R_k = I$ sit at half-width exactly $d$.

`tridiag_structure_scope.py` pins that block tridiagonality of $H(\lambda)$ is forced by the class rather than assumed, with a negative control: the band sweep returns a wrong verdict on a symmetric matrix that lacks the structure.

`curvature_factor_check.py` pins the curvature convention $\nabla^2_{xx}(f_0 - \sum_j \lambda_j f_j) = 2H(\lambda)$ under the paper's no-$\frac{1}{2}$ convention, and enumerates which downstream uses of $H$ are invariant under positive rescaling.

`band_lemma_scope.py` extends the band-$LDL^\top$ soundness and zero-pivot checks to general bandwidth, including even $b$, which is the generality at which the paper states its band lemma.

`band_opcount.py` instruments the shipped band sweep and pins its $O(nb^2) = O(Nd^3)$ operation count against its loop bounds. It is the band sweep's count that is measured; the bordered sweep's count is read off its loop bounds, as the paper's bordered-extension remark states.

`wdd_row_reduction_check.py` pins the identity-coupled row shapes of the paper's diagonal-dominance lemma and the reduction of WDD to the local test $w_k \geq \lambda_k$; the checks cover the singular boundary, including the case where the guarded pivot recursion is silent but the local test is not.

`riccati_margin.py` demonstrates that WDD is sufficient but not necessary for the PSD verdict: on the identity-coupled chain, PSD persists below the WDD threshold $c = 2$, down to $2\cos(\pi/(N+1))$.

`ya_scope.py` pins the boundary of the demonstration family's growth behaviour: its mild pivot growth and its compact recomputed value are properties of the family's shared transition matrix. Swapping the shared CW matrix for the per-stage Yamanaka–Ankersen transition matrix, with everything else unchanged, reproduces the CW pivot sequence bit for bit at $e = 0$. At the probed eccentricities $e \in \{0.1, 0.3, 0.5\}$, with the stage matrices rationalized to the CW family's realized per-entry fidelity, the pivots grow at $\approx 1000$–$1050$ bits per stage against CW's $\approx 778$, and $\mathrm{bit}(t^\star)$ grows at $\approx 1000$ bits per stage instead of staying horizon-bounded. The penalty is already present at $e = 10^{-4}$: the driver is the transition matrix's per-stage time-variance, not the eccentricity magnitude.

**The boundary receipts.** Five receipts pin the fixed-precision boundary analysis of the paper's related-work discussion.

`claim269_recheck.py`**,** `claim305_recheck.py` **and** `claim305_deepchain.py` pin the positive-margin case: pivots away from the semidefinite boundary may round, their positive-width enclosures absorbed by their positive margins.

`claim305_recheck2.py` pins the exactly-computed-zero-pivot scope of the Cholesky clause: LAPACK's `dpotrf` declines the factorization of the singular $H$ with rows $(1, -1)$ and $(-1, 1)$. That an $LDL^\top$ and the symmetric eigensolver are exact there is pinned by `attack3_scope.py` and `claim269_recheck.py`.

`attack3_scope.py` pins the floating-point transcription the paper's conclusion reports — the IEEE-754 double sweep refusing `two_keepout_stage.py`'s certificate at a terminal corner of $-4.44 \times 10^{-16}$ — and the zero-pivot branch on which the bordered and band-$H$ pivot sequences diverge.

# 7 Mechanization of the published note's theorems

The exact S-procedure bracket and the rational $LDL^\top$ PSD test for single primitives are established in the published Podium technical note, *Exact-Rational Certificates in Podium: Constructions, Proofs, and Prior Art* [3]. Its soundness core is mechanized in `PodiumNote.lean`, which sits in a `note/` subdirectory beside the eighteen top-level files and is unchanged since commit `91fb1fa` of the same tree. The regression script of §2 type-checks the file as its nineteenth entry, under the same toolchain pin, and surfaces the axiom audit for `note_thm_sound`, `note_thm_multi` and `note_lem_schur`. The declarations are stated over any linear ordered field, like the corpus's, and mechanize the note's Theorem 1, the lower-bound leg of its Theorem 4 with the accompanying bracket, and the scalar case of its Schur-complement lemma.

`note_thm_sound` (the note's Theorem 1, the lower bound, at one constraint): if $\lambda \geq 0$ and the minorant condition $t \leq f_0(x) - \lambda f_1(x)$ holds at every point $x$, then $t \leq f_0(x)$ at every $x$ with $f_1(x) \geq 0$, hence $t \leq J^\star$. The minorant condition is where $M(\lambda, t) \succeq 0$ enters: the published note derives it from PSD of the bordered matrix by the lifting expansion, and the declaration takes the minorant itself as its hypothesis. Within the corpus, the same step is carried at general order by `general_sprocedure_sound`'s form-identity hypothesis (§4).

`note_thm_multi` (the note's Theorem 4, the certified duality gap, at any number $m$ of constraints): if every $\lambda_i \geq 0$ and $t \leq f_0(x) - \sum_i \lambda_i f_i(x)$ at every point, then $t \leq f_0(x)$ at every $x$ with all $f_i(x) \geq 0$ — the theorem's opening statement, that Theorem 1's minorant argument uses nothing of $m = 1$. The declaration is stated at one fixed $(\lambda, t)$; the theorem's supremum characterization of the Shor value $d^\star$, and the strictness of the gap for $m \geq 2$, are outside it.

`note_bracket` (the same theorem's closing observation): a lower bound $t \leq f_0$ over a feasible set, supplied as a hypothesis, together with a feasible witness $\bar{x}$, yields $t \leq f_0(\bar{x})$ and, at every feasible $x$, $f_0(\bar{x}) - (f_0(\bar{x}) - t) \leq f_0(x)$ — the form in which the width $f_0(\bar{x}) - t$ of the bracket $[t, f_0(\bar{x})]$ bounds the suboptimality of $\bar{x}$. Feasibility enters as an abstract predicate, so the statement is independent of the constraint form.

`note_lem_schur` (the note's Schur-complement lemma, scalar case): for $a > 0$, the bordered matrix $M = \left[\begin{smallmatrix} a & b/2 \\ b/2 & c-t \end{smallmatrix}\right]$ is PSD in the quantifier sense $\forall v,\ 0 \leq v^\top M v$ if and only if $t \leq c - b^2/(4a)$; the right-hand side is the dual value $\inf_x(ax^2 + bx + c)$. The equivalence is mechanized in both directions. The published lemma is stated for a matrix block $A(\lambda) \succ 0$ with $g(\lambda) = c - \frac{1}{4} b^\top A(\lambda)^{-1} b$; the declaration is its $1 \times 1$ instance.

## 8 Scope: what the theorems take as hypotheses

The shipped sweeps are not the mechanized object: the corpus proves the acceptance predicate's equivalence with positive semidefiniteness. The claim that the concrete band elimination in `podium.verify.riccati` computes such a factorization is a refinement hypothesis, taken rather than proved. The same holds for the dense reference $M$-test of `podium.verify.barrier`, which pins the band elimination and against which every emitted certificate is checked.

Three things are outside the mechanized perimeter: the border-carrying sweep; the band-restriction equality (band pivots equal dense pivots); and the operation count. The paper pins the first by test against the dense $M$-test and reads the count off the shipped loop bounds; the corpus contains none of the three.

The form identity is discharged per instance: `general_sprocedure_sound`'s third hypothesis and the capstone's matching hypothesis are proved by `ring` for each concrete assembly, never once in general.

Lagrangian minimality is supplied from outside the corpus: `certified_global_optimum` takes it as a hypothesis, and the quadratic-convexity step that produces it in the paper's proof is not mechanized.

The bridge is compiled separately: `PosSemidefBridge.lean`, `LDLTSound.lean` and `SchurStep.lean` each declare their own top-level `IsPSD` under the same unnamespaced name, and the three files import none of one another. So no compiled artifact links the bridge to the predicate the acceptance lemmas decide. The three formulas are definitionally identical over $\mathbb{Q}$; the match is read off by inspection.

The second leg of the paper's diagonal-dominance lemma is the Schur-complement preservation of weak diagonal dominance, resting on Carlson–Markham [1] together with the limiting argument in that proof. That leg is outside the corpus; only the Gershgorin leg is machine-checked.

## References


[1] David Carlson and Thomas L. Markham. Schur complements of diagonally dominant matrices. *Czechoslovak Mathematical Journal*, 29(2):246–251, 1979.

[2] Adi Oltean. Certifying trajectory optimality in exact arithmetic: An exact-rational Riccati certificate for penalty-coupled trajectory QCQPs, 2026. Companion paper; distributed with this note.

[3] Adi Oltean. Exact-rational certificates in podium: Constructions, proofs, and prior art, 2026. Technical note, Zenodo, `https://doi.org/10.5281/zenodo.21247380`.